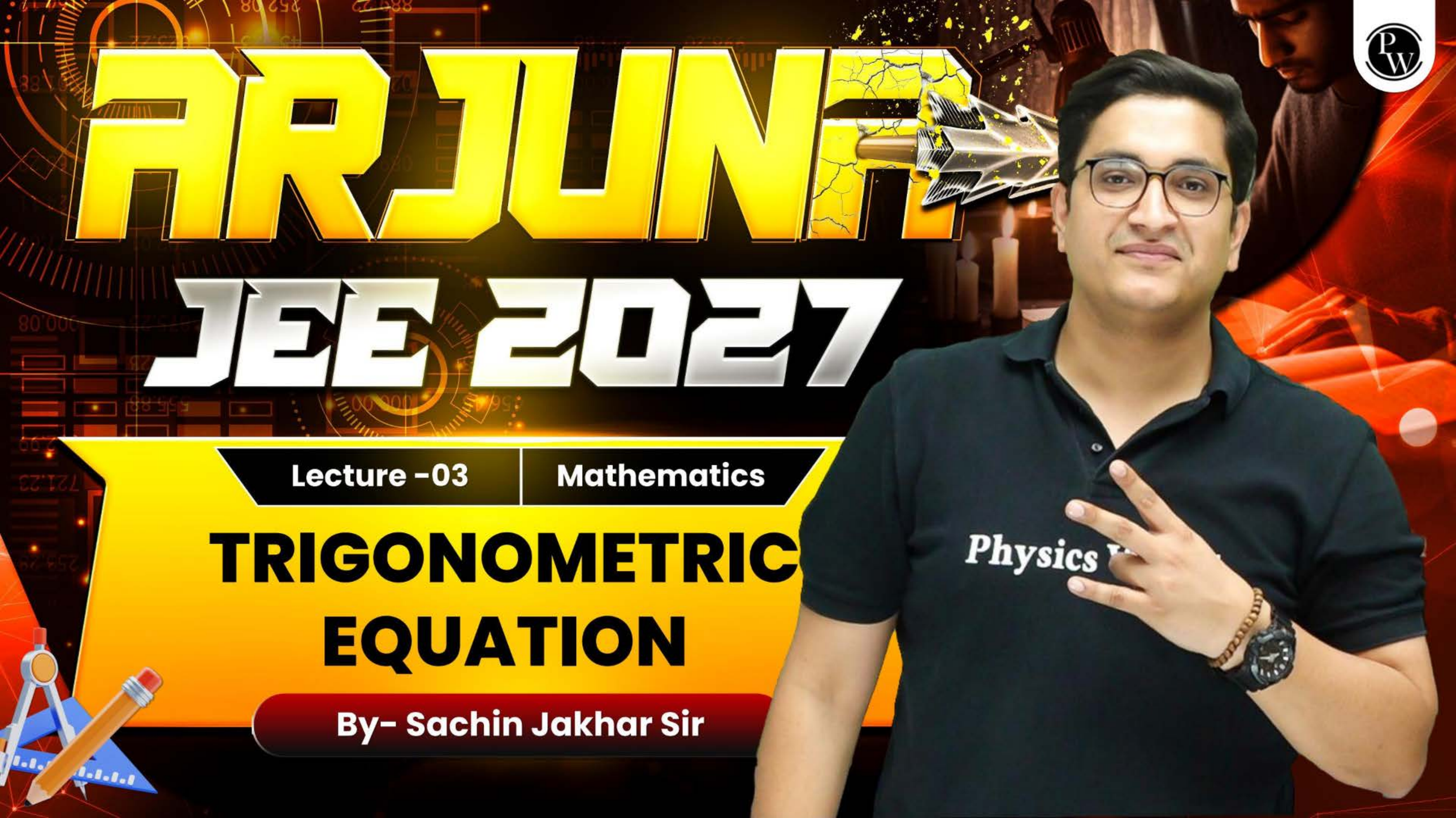
ARJUNA JEE 2027

Lecture -03 | Mathematics

TRIGONOMETRIC EQUATION

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Topics *to be covered*

A **Types of Trigonometric Equations**

B **Question Practice**

C

D



H.W. Discussion

Acidic Question

84%

If $y=\frac{17+5\sin x+12\cos x}{17-5\sin x-12\cos x}$, then the minimum and maximum values of y are respectively

A $\frac{1}{15}, 15$

B $\frac{2}{15}, \frac{15}{2}$

C $\frac{1}{8}, 8$

D $\frac{4}{17}, \frac{17}{4}$

$\sqrt{5^2+12^2}=13$.

say

$5\sin x+12\cos x=\alpha \in [-13,13]$

Simplify

$y=\frac{17+\alpha}{17-\alpha}=-\left(\frac{17-\alpha-34}{17-\alpha}\right)=-\left(1-\frac{34}{17-\alpha}\right)=\boxed{\frac{34}{17-\alpha}-1}$

$\Rightarrow$ C & d

$\boxed{\frac{y+1}{y-1}=\frac{17+\alpha+17-\alpha}{17+\alpha-17+\alpha}=\frac{34}{2\alpha}=\frac{17}{\alpha}}$

$-\alpha \in [-13,13]$

$17-\alpha \in [4,30]$

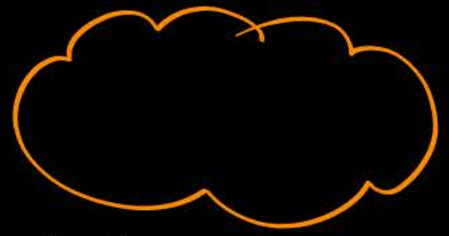
$\frac{1}{17-\alpha} \in \left[\frac{1}{30},\frac{1}{4}\right]$

$\frac{34}{17-\alpha} \in \left[\frac{34}{30},\frac{34}{4}\right]=\left[\frac{17}{15},\frac{17}{2}\right]$

-1

$\left[\frac{2}{15},\frac{15}{2}\right]$

Acidic Question



84%

If $y=\frac{17+5\sin x+12\cos x}{17-5\sin x-12\cos x}$, then the minimum and maximum values of y are respectively

A $\frac{1}{15}, 15$

B $\frac{2}{15}, \frac{15}{2}$

C $\frac{1}{8}, 8$

D $\frac{4}{17}, \frac{17}{4}$

say

$5\sin x + 12\cos x = \alpha \in [-13, 13]$

$N^r \in [4, 30]$

$D^r \in [4, 30]$

$\Rightarrow y_{max.} = \frac{N^r_{max.}}{D^r_{min.}} = \frac{30}{4} = \frac{15}{2}$

(#) $y = \dfrac{1 + 2\cos x}{1 - 2\cos x}$ → $[-1, 3]$ → $[-1, 3]$

$y_{max.} = \dfrac{3}{-1} = -3$

$y_{min.} = \dfrac{-1}{3}$

X

(#) $y = \dfrac{1+2\cos x}{1-2\cos x}$ → $[-1,3]$ → $[-1,3]$

$$y = \frac{1+2\alpha}{1-2\alpha} = \frac{1-(-2\alpha)}{1-2\alpha} = \frac{2-(1-2\alpha)}{1-2\alpha} = \frac{2}{1-2\cos x} - 1$$

$-2\cos x \in [-2,2]$

$1-2\cos x \in [-1,3]$

$\dfrac{2}{1-2\cos x} \in (-\infty,-2] \cup \left[\frac{2}{3},\infty\right)$

$\dfrac{2}{1-2\cos x} - 1 \in (-\infty,-3] \cup \left[\frac{-1}{3},\infty\right)$

(#) $y = \dfrac{1 + 2\cos x}{1 - 2\cos x}$

$y - 2\cos x \cdot y = 1 + 2\cos x$

$y - 1 = 2\cos x + 2\cos x\, y$

$y - 1 = 2\cos x\,(1 + y)$

$\dfrac{1}{2}\left(\dfrac{y-1}{y+1}\right) = \cos x$

$\cos x \in [-1, 1]$

$-1 \le \dfrac{1}{2}\left(\dfrac{y-1}{y+1}\right) \le 1$

double ineq., solve for y

Critical Thinking Question

$y = 18\sec^2 x + 8\cos^2 x$, find minimum value of y & also find corresponding value of x.

"Reciprocal"

$18\sec^2 x$, $8\cos^2 x \rightarrow AM \geq GM$

$$\frac{18\sec^2 x + 8\cos^2 x}{2} \geq \left(18\sec^2 x \cdot 8\cos^2 x\right)^{\frac{1}{2}}$$

$18\sec^2 x + 8\cos^2 x \geq 12 \times 2$

24 ✗ wrong.

$\cos x \in [-1, 1]$

$\cos^2 x \in [0, 1]$

• equality hold

$18\sec^2 x = 8\cos^2 x$

$\frac{18}{\cos^2 x} = 8\cos^2 x$

$\frac{9}{4} = \frac{18}{8} = \cos^4 x$

$\frac{9}{4}$ greater than 1.

$\cos^4 x$ → $[0, 1]$

$y = 18\sec^2 x + 8\cos^2 x$, find minimum value of y & also find corresponding value of x.

"Reciprocal"

$$y = \underbrace{10\sec^2 x}_{\text{min. value} = 10 \text{ when } x=0} + \underbrace{8\sec^2 x + 8\cos^2 x}_{\text{AM} \geq \text{GM},\ \text{min.} = 16 \text{ when } (x=0)}$$

$$\Big|_{\text{min.}} = 10 + 16 = 26 \quad (\text{when } x = 0)$$

$8\sec^2 x + 8\cos^2 x \geq 16$

when:

$8\sec^2 x = 8\cos^2 x$

$1 = \cos^4 x$

$\cos x = \pm 1$

$x = 0, \pi, 2\pi, 3\pi \ldots$

Ques:

$\left(2^{x}+2^{-x}\right), \left(f(x)\right), \left(3^{1-x}+3^{1+x}\right)$ are in A.P.

Then find min. value of $f(x)$.

Hw.

Question [IIT-JEE-2005]

Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then

$(\frac{1}{2})^{\frac{1}{2}}$ $(\frac{1}{2})^{2}$ $2^{\frac{1}{2}}$ 2^{2}

A $t_1 > t_2 > t_3 > t_4$

B $t_4 > t_3 > t_1 > t_2$

C $t_3 > t_1 > t_2 > t_4$

D $t_2 > t_3 > t_1 > t_4$

$\theta \in \left(0, \frac{\pi}{4}\right)$

$\cot\theta > 1$

$\tan\theta < 1$

say: $\cot\theta = 2$ & $\tan\theta = \frac{1}{2}$

$t_4 > t_3 > t_1 > t_2$

A B C D

\# $\left(\frac{1}{2}\right)^2$ $(2)^2$ $(2)^{\frac{1}{2}}$ $\left(\frac{1}{2}\right)^{\frac{1}{2}}$

$A < D < C < B$

Question [JEE Mains-2021]

$\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$ & $\sin\frac{3\pi}{4} = +\frac{1}{\sqrt{2}}$

The range of the function

$f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}-x\right) - \cos\left(\frac{3\pi}{4}-x\right)\right)$ is

A $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$

B $[0, 2]$

C $(0, \sqrt{5})$

D $[-2, 2]$

$\cos\frac{3\pi}{4}\cos x - \sin\frac{3\pi}{4}\sin x + \cos\frac{\pi}{4}\cdot\cos x - \sin\frac{\pi}{4}\sin x + \cos\frac{\pi}{4}\cdot\cos x + \sin\frac{\pi}{4}\sin x$

$-\cos\frac{3\pi}{4}\cos x - \sin\frac{3\pi}{4}\sin x$

$\log_{\sqrt{5}}([1,5])$

$[0,2]$

$2\times\frac{1}{\sqrt{2}}\cos x - 2\times\frac{1}{\sqrt{2}}\sin x$

$\sqrt{2}(\cos x - \sin x)$

$\sqrt{2}\times[-\sqrt{2},\sqrt{2}]$

$[-2,2]$

TRIGONOMETRIC EQUATION

Question [JEE Mains-2021]

The number of solutions of the equation $|\cot x| = \cot x + \frac{1}{\sin x}$ in the interval $[0, 2\pi]$ is

$x \in$ I OR III quad.

$\cot x \rightarrow +ve$.

$\cot x = \cot x + \frac{1}{\sin x}$.

$0 = \frac{1}{\sin x}$

$\downarrow$

ϕ.

$x \in$ II OR IV quad.

$\cot x \rightarrow$ (-ve)

$-\cot x = \cot x + \frac{1}{\sin x}$

$-2\cot x = \frac{1}{\sin x}$

$-\frac{2\cos x}{\sin x} = \frac{1}{\sin x} \Rightarrow \cos x = -\frac{1}{2}$

II quad.

NOS = 2 ✗

NOS = 1 ✓

Question [JEE Mains-2020]

The number of distinct solutions of the equation, $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0, 2\pi]$, is



LAST CLASS RECAP

(1) $\sin\theta = \sin\alpha \Rightarrow \boxed{\theta = n\pi + (-1)^n \alpha}$

(2) $\cos\theta = \cos\alpha \Rightarrow \boxed{\theta = 2n\pi \pm \alpha}$

(3) $\tan\theta = \tan\alpha \Rightarrow \boxed{\theta = n\pi + \alpha}$

(4) $\left.\begin{matrix}\sin^2\theta = \sin^2\alpha \\ \cos^2\theta = \cos^2\alpha \\ \tan^2\theta = \tan^2\alpha\end{matrix}\right\} \Rightarrow \boxed{\theta = n\pi \pm \alpha}$

$n \in I$.

Question

The general solution of the following equation : $\cos 3\theta = -\frac{1}{2}$

A $2n\pi \pm \frac{2\pi}{3}$

B $2n\pi \pm \frac{\pi}{3}$

C $\frac{2n\pi}{3} \pm \frac{2\pi}{9}$

D $\frac{2n\pi}{3} \pm \frac{4\pi}{3}$

θ α

$\cos 3\theta = \cos \frac{2\pi}{3}$

$3\theta = 2n\pi \pm \frac{2\pi}{3}$

$\theta = \frac{2n\pi}{3} \pm \frac{2\pi}{9}$, $n \in I$.

Question

The general solution of $\tan\left(\frac{2}{3}\theta\right) = \sqrt{3}$.

A $\frac{3n\pi}{2} + \frac{\pi}{2}; n \in I$

B $\frac{n\pi}{2} + \frac{\pi}{2}; n \in I$

C $n\pi \pm \frac{\pi}{2}; n \in I$

D None of these

$\tan\left(\frac{2\theta}{3}\right) = \tan\frac{\pi}{3}$ (θ, α)

$\frac{2\theta}{3} = n\pi + \frac{\pi}{3}$

$\times \frac{3}{2}$

$\theta = \frac{3n\pi}{2} + \frac{\pi}{2}$

Question

Page No.55

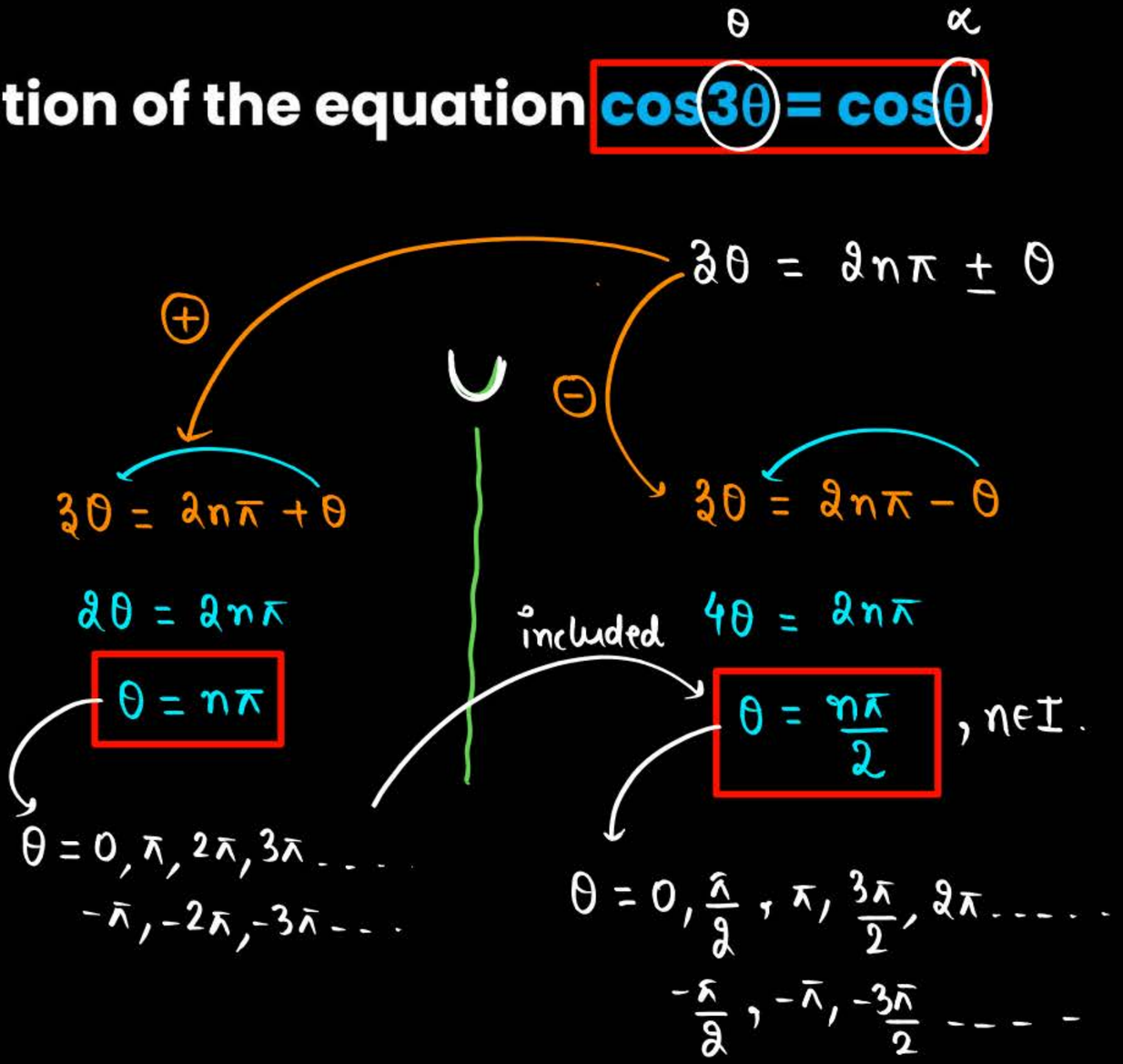
Complete set of general solution of the equation $\cos 3\theta = \cos\theta$.

(A) $\theta \in \{n\pi\}, n \in I$

(B) $\theta \in \left\{(2n+1)\frac{\pi}{2}\right\}, n \in I$

(C) $\theta \in \left\{\frac{n\pi}{2}\right\}, n \in I$

(D) $\theta \in \left\{(4n+1)\frac{\pi}{2}\right\}, n \in I$



Acidic Question

Page No.56

Complete set of general solution of $\sin 9\theta = \sin\theta$.

(θ and α)

A $\theta \in \left\{(2m+1)\frac{\pi}{10}\right\}, m \in I$

B $\theta \in \left\{\frac{m\pi}{4}\right\}, m \in I$

C $\theta \in \left\{\frac{m\pi}{4}\right\} \cup \left\{(2m+1)\frac{\pi}{10}\right\}, m \in I$

D $\theta = n\pi, n \in I$

$9\theta = n\pi + (-1)^n \theta$

\# $n \equiv$ even integer. OR $n = 2m,\ m \in I$.

$(-1)^n = (-1)^{2m} = ((-1)^2)^m = 1$

$9\theta = 2m\pi + \theta$

$8\theta = 2m\pi$

$\theta = \frac{2m\pi}{8}$

$\theta = \frac{m\pi}{4}, m \in I.$

$$9\underline{\theta} = \underline{n}\pi + (-1)^n \underline{\theta}$$

\# $n \equiv$ even integer. OR $n = 2m,\ m \in I$.

$$(-1)^n = (-1)^{2m} = ((-1)^2)^m = 1$$

$$9\theta = 2m\pi + \theta$$

$$8\theta = 2m\pi$$

$$\theta = \frac{2m\pi}{8}$$

$$\theta = \frac{m\pi}{4},\ m \in I$$

$$0, \frac{\pi}{4}, \frac{2\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{6\pi}{4}$$

$\frac{2\pi}{4} \to \frac{\pi}{2}$, $\frac{6\pi}{4} \to \frac{3\pi}{2}$

union

$$9\underline{\theta} = \underline{n}\pi + (-1)^n \underline{\theta}$$

$n =$ odd integer

OR

$n = (2m+1),\ m \in I$

$$(-1)^n = (-1)^{2m+1} = -1.$$

$$9\theta = (2m+1)\pi - \theta$$

$$10\theta = (2m+1)\pi$$

$$\theta = (2m+1)\frac{\pi}{10},\ m \in I$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}, \frac{11\pi}{10}, \frac{15\pi}{10}$$

$\frac{15\pi}{10} \to \frac{3\pi}{2}$

Question

Page No.56

Solve following equation: $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

$(\quad)^2 + (\quad)^2 = 0$

$\sin\theta = \frac{\sqrt{3}}{2}$ $\qquad$ $\tan\theta = -\sqrt{3}$

$\sin\theta = \sin\frac{\pi}{3}$ $\qquad$ $\tan\theta = \tan\left(-\frac{\pi}{3}\right)$

$\theta = n\pi + (-1)^n \frac{\pi}{3}$ $\qquad$ $\theta = n\pi + \left(-\frac{\pi}{3}\right)$

odd multiple of $\pi - \frac{\pi}{3}$

OR

$\theta = (2n+1)\pi - \frac{\pi}{3} = 2n\pi + \frac{2\pi}{3}$

$\frac{\pi}{3}, \pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, \ldots$

$-\pi - \frac{\pi}{3}, -2\pi + \frac{\pi}{3}, \ldots$

$-\frac{\pi}{3}, \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, \ldots$

$-\pi - \frac{\pi}{3}, -2\pi - \frac{\pi}{3}, \ldots$

Question

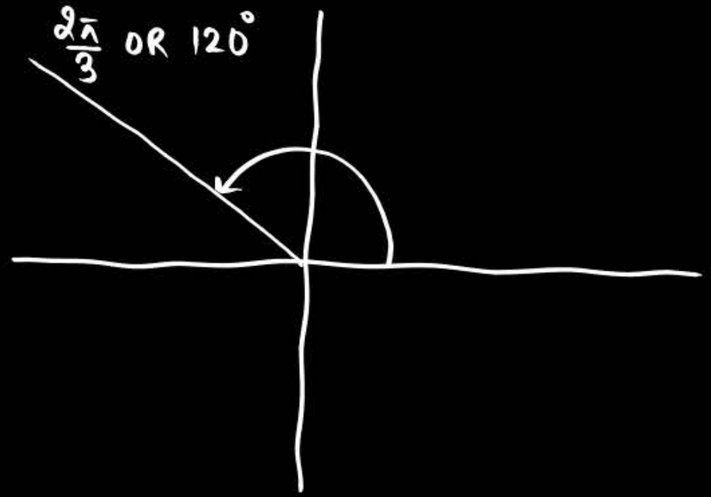
Page No.56

Solve following equation: $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

$(\underset{\geq 0}{\quad})^2 + (\underset{\geq 0}{\quad})^2 = 0$

M-2

$\sin\theta = \frac{\sqrt{3}}{2}$ $\cap$ $\tan\theta = -\sqrt{3}$



$\theta \rightarrow$ II quad.

$\theta = \frac{2\pi}{3}$

Generalise (add $2n\pi$)

$2\pi, 4\pi, 6\pi$
$-2\pi, -4\pi \ldots$

$\theta = 2n\pi + \frac{2\pi}{3}$, $n \in I$.

RAPID FIRE

Solutions in case if two equations are given:

1. If $\sin(\theta) = \frac{1}{\sqrt{2}}$ and $\cos(\theta) = -\frac{1}{\sqrt{2}}$, then find the general values of θ.
2. If $\sin(\theta) = \frac{1}{\sqrt{2}}$ and $\tan(\theta) = -1$, then find the general values of θ.
3. If $\cos\theta = \frac{1}{\sqrt{2}}$ and $\tan\theta = -1$, then find the general value of θ.

Same Q.

θ = II quad.

$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$\theta \in \left\{2n\pi + \frac{3\pi}{4}\right\}, n \in I$.

Question

If $\cos\theta = -\frac{1}{\sqrt{2}}$ and $\tan\theta = 1$, then the general value of θ is



A $2n\pi + \frac{\pi}{4}$

B $(2n+1)\pi + \frac{\pi}{4}$

C $n\pi + \frac{\pi}{4}$

D $n\pi \pm \frac{\pi}{4}$

[where n ∈ I],



DHYAAN RKHNA!!

1. **Avoid squaring in case of squaring final solution should be checked.**

   Squaring se extra solutions aa skte hai.

2. **Avoid cancellation in multiplication.**

   Cancellation leads to loss of original solutions.

$\Rightarrow ab = bc \Rightarrow ab - bc = 0 \Rightarrow b(a-c) = 0$ → $b = 0$ OR $a = c$.

$\Rightarrow \frac{b}{a} = \frac{c}{a} \Rightarrow b = c$

$\Rightarrow a + b = a + c \Rightarrow b = c$

DHYAAN RKHNA!!

3. Answer shouldn't contain values for which any of given function is not defined.

***Ex.* : Solve : $\tan 3x - \tan 2x = 1 + \tan 3x \tan 2x$**

$$\frac{\overset{A}{\tan 3x} - \overset{B}{\tan 2x}}{1 + \underset{A}{\tan 3x} \cdot \underset{B}{\tan 2x}} = 1.$$

$\tan(3x - 2x) = 1.$

$\tan x = \tan \frac{\pi}{4}$

$x = n\pi + \frac{\pi}{4}, n \in I.$ (X)

$x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{-3\pi}{4} \ldots$

ϕ (No sol.^{n})

$\left.\begin{matrix}\tan x \\ \sec x\end{matrix}\right\} \Rightarrow x \neq (2n+1)\frac{\pi}{2}$

$\left.\begin{matrix}\cot x \\ \operatorname{cosec} x\end{matrix}\right\} \Rightarrow x \neq n\pi$

Question

Solve : $\tan x \tan 4x = 1$

$$\frac{\sin x \cdot \sin 4x}{\cos x \cdot \cos 4x} = 1$$

$$0 = \cos x \cdot \cos 4x - \sin x \cdot \sin 4x$$

$$\# \quad \cos\frac{\pi}{2} = \cos(5x)$$

$$5x = 2n\pi \pm \frac{\pi}{2} = \frac{4n\pi \pm \pi}{2}$$

$$x = (4n \pm 1)\frac{\pi}{10}, \quad n \in I$$

$$\frac{\tan 4x}{\cot x} = 1$$

$$\tan 4x = \cot x \quad \alpha$$

$$\tan(4x) = \tan\left(\frac{\pi}{2} - x\right)$$

$$4x = n\pi + \frac{\pi}{2} - x$$

$$5x = \frac{2n\pi + \pi}{2} = (2n+1)\frac{\pi}{2}$$

$$x = (2n+1)\frac{\pi}{10}, \quad n \in I$$

$5x = 2n\pi \pm \frac{\pi}{2} = \frac{4n\pi \pm \pi}{2}$

$x = (4n \pm 1)\frac{\pi}{10}, n \in I$

exactly same.

$x = (2n+1)\frac{\pi}{10}, n \in I$

$n = -3$ $\quad$ $n = 2$ $\quad$ $n = 7$

$-\frac{5\pi}{10}, -\frac{3\pi}{10}, -\frac{\pi}{10}, \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}, \frac{11\pi}{10}, \frac{13\pi}{10}, \frac{15\pi}{10} \ldots$

$\frac{5\pi}{10} = \frac{\pi}{2}$

$\tan x \cdot \tan 4x = 1$

$x \neq \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \ldots$

$4x \neq \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \ldots$

$x \neq \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8} \ldots$

① # $x \in (2n+1)\frac{\pi}{10} - \left\{(2n+1)\frac{\pi}{2}\right\}$ ✓

$n = 12$ $\quad \frac{25\pi}{10} = \frac{5\pi}{2}$ $\quad \frac{3\pi}{2}$

② $x \in (2n+1)\frac{\pi}{10}, n \neq 2, 7, 12, 17, 22, \ldots$

G.T. $= 2 + (m-1)5$
$= 5m - 3$

③ $x \in (2n+1)\frac{\pi}{10}, n \neq 5m-3, m \in I.$

Question

If $\tan\theta = \cot 2\theta$, then the value of θ is



A $(2n+1)\frac{\pi}{6}; n \in I$

B $(2n+1)\frac{\pi}{6}; n \in I; n \neq 3m+1; m \in I$

C $(2n+1)\frac{\pi}{6}; n \in I; n \neq 2m+1; m \in I$

D none

Question [JEE (Adv.)-2018]

Let a, b, c be three non zero numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c,\ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is



DO IT BY YOURSELF (DIBY)

DIBY-01

Solve : $\sqrt{3}\sec 2\theta = 2$.

[Ans. : $\theta = n\pi \pm \frac{\pi}{12}, n \in I$]

DIBY-02

Find general solution of equation $\cot x = \sqrt{3}$

(A) $x = n\pi + \frac{\pi}{3}$

(B) $x = n\pi + \frac{\pi}{6}$

(C) $x = n\pi + \frac{2\pi}{3}$

(D) $x = \frac{9\pi}{6}\frac{5\pi}{6}$

[Ans. : (B)]

DIBY-03

The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is:

(A) 0 (B) 2 (C) 6 (D) 8

[Ans. : (C)]

DIBY-04

General solution of the equation $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

[Ans. : $(6n+2)\frac{\pi}{3}, n \in I$]

DIBY-05

Find the principal solution of $\sin 2x = -\frac{1}{2}$

(A) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(C) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(D) None of these

[Ans. : (C)]

DIBY-06

Find the number of solution $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$.

[Ans. : 05]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians